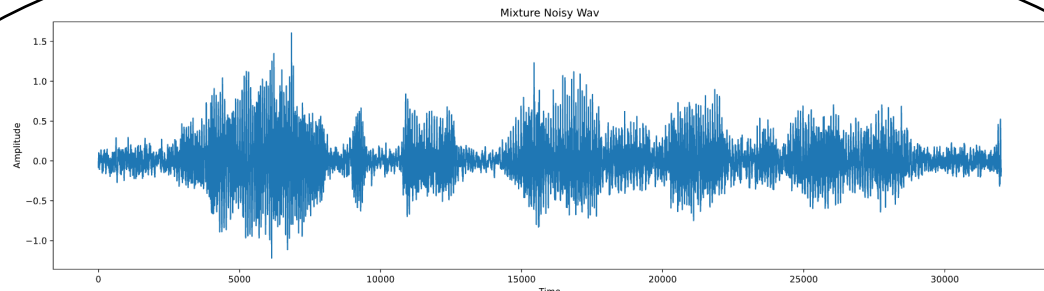
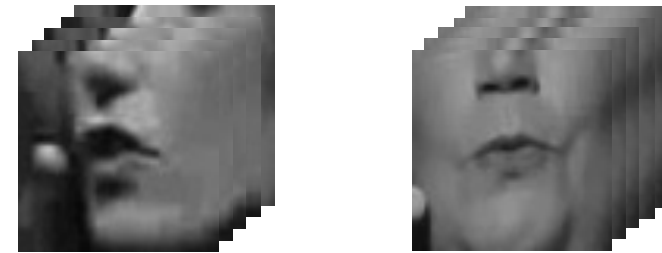


Modality 1:
The embedded mixture of
audio records from two
speakers with additional noise



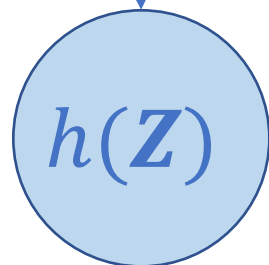
$$S = \text{Embed}(S_1, S_2)$$

Modality 2:
The embedded video
frames of the cropped lip
region from two speakers

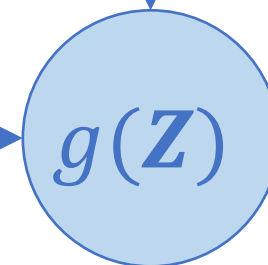


$$V = \text{Embed}(V_1, V_2)$$

The audio
latent features
for the two
speakers



The facial
latent features
for the two
speakers



?

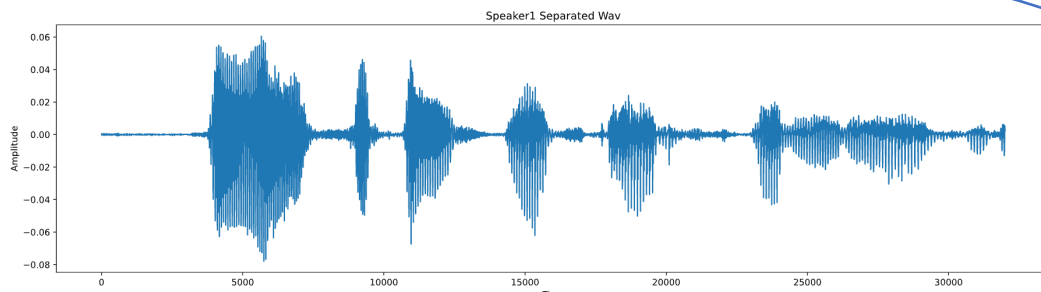
?

Z

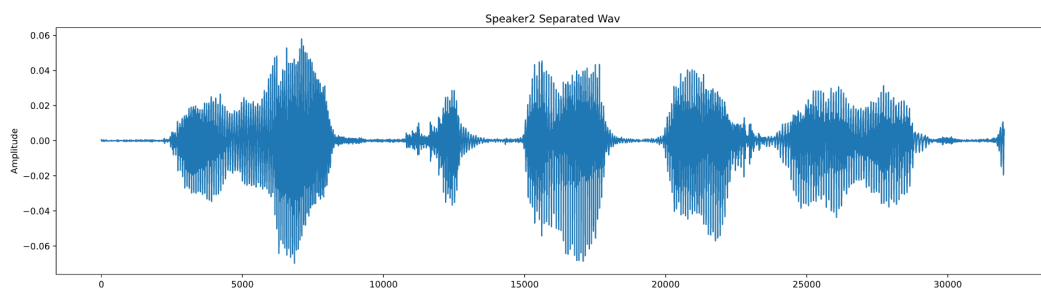
Z_1

Z_2

$\hat{S}_1$



$\hat{S}_2$



Target:
The separated and denoised
audio records of the two
speakers from the mixture of
audio records